Choosing the two flip angles of an MPME protocol: an optimal-design study

Technical note

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October 22, 2026

Abstract

The multi-pathway multi-echo (MPME) protocol of Cheng et al. [1] acquires two steady-state scans that differ only in the flip angle (15° and 330°). We ask which pair of flip angles best determines T_1 and the other parameters. Using the Ernst-angle form of the signal invariants, we show that each scan reports only the offset of its flip angle from the tissue's Ernst angle, that the information of a design is the sum of two single-scan informations, and that two scans determine B_1^+ through a lever $c(x_2) - c(x_1)$, $c(x) = x/\sin x$. This gives exact closed-form Cramér–Rao bounds and a closed-form bias for pulse-ratio errors. An exhaustive search over 145 530 pairs, the transmit range $B_1^+ \in [0.7, 1.3]$ and five tissues, with an analytic check for exact aliases, finds a flat optimum that contains the published design: no choice of angles improves the mean T_1 bound by more than 2% or the worst case by more than 3.2%. The angles matter instead for robustness. Unmodelled finite pulses and flip-angle spreads bias T_1 most where the scan-2 pulse approaches 360°. A second pulse near 270°, which keeps that crossing out of the B_1^+ range, nearly halves the worst-case B_1^+ bound, needs 30–33% less energy at equal pulse duration, has no aliases when fitted within its crossing-free range, and at $B_1^+ = 1$ is nine times less biased in WM and GM by unmodelled finite pulses of the same peak B_1^+ (but not at the top of the B_1^+ range), at a 3–7% cost in mean T_1 precision. Every design transfers a pulse-ratio error to T_1 with a factor of at least 1.6, and every design should model its finite pulses.

Summary of findings

1. **Structure (Section 2).** A design's information is $\mathcal{I}_1(B_1^+\alpha_1) + \mathcal{I}_1(B_1^+\alpha_2)$. Per scan, the T_1 column of the Jacobian is $h(J_{B_1^+}/c - J_{M_0})$, so a scan sees B_1^+ and T_1 only through its offset from the Ernst angle. Pulses with equal $|\tan(x/2)|$ (e.g. 30° and 330°) carry the same per-scan information and differ only in the rate $c(x) = x/\sin x$ at which they move with B_1^+ (1.05 and −11.5).

2. **Closed forms (Propositions 3 and 4).** $\mathrm{SD}(\ln B_1^+)$ is the SD of the difference of the two offsets divided by the lever $L = c_2 - c_1$, and $\mathrm{SD}(\ln T_1)$ that of a lever-weighted combination; the B_1^+ uncertainty reaches T_1 with the factor $c_1/h \approx 2$. A pulse-ratio error ε gives $\delta \ln T_1 = -c_1 c_2 \varepsilon/(hL)$, $|\delta \ln T_1| \geq 1.64\,\varepsilon$, with zero residual.

3. **Flat optimum (Section 3).** Best mean T_1 bound 49.2 at 14°/309° (published: 50.2); best worst case 100.3 at 20°/276° (published: 103.6). The 5% band extends over α_1 = 11–20°, α_2 = 267–348°.

4. **The 360° crossing.** Where $B_1^+\alpha_2 = 360°$ scan 2 measures only B_1^+; the T_1 bound peaks there (0.5% wide in B_1^+ for CSF, 1–1.3% for parenchyma), and this peak decides the worst case.

5. **Aliases (Section 4).** Exact aliases are the other roots of $\tan(B_1^+\alpha_2/2)/\tan(B_1^+\alpha_1/2) = \xi_2/\xi_1$ and depend on B_1^+ alone; $\mathrm{d}\ln|f|/\mathrm{d}\ln B_1^+$ equals the lever. A design whose pulses stay within $(0°, 180°)$ and $(180°, 360°)$ over the fitting range has none.

6. **Constraints (Section 5).** Pulses above $\approx 300°$ buy nothing; limiting the largest angle to 240° costs 6–14%, to 180° 25–52%. A second pulse below 90° removes the lever: 15°/85° has a 6.8 times larger worst-case T_1 bound (4.5 times with $\alpha_1 = 34°$) for 7% of the scan-2 energy. For the two-stage pipeline $\alpha_1 = 17$–$19°$ is best.

7. **Model error (Section 6).** Unmodelled finite pulses (equal peak B_1^+, 1 ms for 330°) bias T_1 by 10–20% wherever the scan-2 angle passes 330–350° (20% at $B_1^+ = 1$ with 330°, 2% with 270°); slab-edge flip-angle spreads near or across 360° bias T_1 by 13–74%. On resonance both are often hard to detect. Recommendation (Section 7): $\alpha_1 \approx 17$–$21°$, $\alpha_2 \approx 255$–$270°$ for $B_1^+ \le 1.3$, finite pulses in the model, a calibrated pulse ratio.

1 Setting and design problem

1.1 The model in brief

The MPME sequence [1] is an unbalanced steady-state gradient echo that records three configuration pathways ($k = +1, 0, -1$), three echoes each, in each of two scans. The two scans share the repetition time $T_R = 25$ ms and the readout and differ only in the nominal flip angle; the actual angle of scan i is $x_i = B_1^+\alpha_i$. The unknowns are $\theta = (\ln M_0, \ln B_1^+, \ln T_1, \ln T_2, \ln T_2^*, \Delta\omega, \phi_0)$, with logarithms for the positive parameters so that standard deviations are relative. We use the following results of the companion technical note [2].

- *Two invariants per scan.* Each pathway amplitude is $a_k = v\,G_k(u, E_2, \Delta\omega)$, where u and v depend on the flip angle and T_1 only, and G_k does not depend on either.

- *Ernst-angle form.* With the Ernst angle $\alpha_E = \arccos \mathrm{e}^{-T_R/T_1}$ and $\zeta = \tan(\alpha_E/2) = \sqrt{\tanh(T_R/2T_1)}$, define the *offset* of a sample from the Ernst angle

$$w = \ln|\tan(x/2)| - \ln\zeta. \tag{1}$$

  Then $u = \tanh w$ and $|v| = \zeta\,\mathrm{sech}\,w$, with $\mathrm{sign}\,v = \mathrm{sign}\tan(x/2)$. On the axis $s = \ln|\tan(x/2)|$ the spoiled-gradient-echo amplitude is a sech bump centred at the Ernst angle; a scan samples it at s.

- *What the data determine.* After the echo decay and phases have fixed T_2, T_2^*, $\Delta\omega$ and ϕ_0, one scan determines its offset w and the amplitude $\ell = \ln(M_0\zeta)$; two scans determine w_1, w_2 and the shared ℓ, three numbers for (M_0, B_1^+, T_1). One scan alone cannot determine B_1^+.

1.2 The design problem

Design variables. The nominal angles $0 < \alpha_1 < \alpha_2 \le 540°$; T_R and the readout are fixed, so every design has the same acquisition time.

Criterion. The Cramér–Rao bound (CRLB) on $\ln T_1$ with all seven parameters free, in units of σ/M_0 (complex Gaussian noise, standard deviation σ per real and imaginary part). Bounds on $\ln B_1^+$, $\ln T_2$ and $\ln M_0$ are reported alongside.

Robustness range. The design must work for every $B_1^+ \in [0.7, 1.3]$ (the brain at 3 T, central to peripheral) and the five tissues of Table 1. We report the *worst case* of the bound over this range and its *mean* (uniform over B_1^+, equal weight per tissue).

Table 1: Tissues of the robustness range ($M_0 = 1$ for all, so bounds are relative to each tissue's own signal). Ernst angles at $T_R = 25$ ms.

Tissue	T_1 (ms)	T_2 (ms)	T_2^* (ms)	α_E (°)
CSF	4000	1500	900	6.4
grey matter (GM)	1350	90	60	11.0
white matter (WM)	850	66	50	13.8
deep GM	1130	58	40	12.0
lesion	1500	120	80	10.4

Pipelines. Two uses of the data lead to different optima.

- *Joint fit*: both scans at full resolution, all parameters fitted per voxel; the criterion is the joint T_1 bound above.
- *Two-stage*: a short, low-resolution scan 2 gives a smooth B_1^+ map whose noise is negligible after smoothing [2, Sec. 5], and T_1 comes from scan 1 with B_1^+ known. The criterion for α_1 is the scan-1 T_1 bound with B_1^+ known; α_2 only needs to determine B_1^+ reliably.

Constraints. The scans are acquired one after the other, so the specific absorption rate (SAR) of each scan scales as the square of its own angle for a fixed pulse, and a peak-B_1^+ limit fixes the shortest duration of the largest pulse. Both constrain the largest angle $\alpha_{\max} = \alpha_2$; we report the best design as a function of $\alpha_{\max}$.

Identifiability. A design must not admit a complex alias (Definition 1) for any true B_1^+ in the robustness range within the fitting range $B_1^+ \in [0.6, 1.4]$; this is checked analytically (Section 4). Magnitude aliases are allowed: complex data, the FID-sign test or, for some designs, a narrower fitting range remove them.

Model error. Finally, the shortlisted designs are compared for their bias under errors the fitting model ignores: a pulse-ratio error, finite pulse duration, slice or slab profile (Section 6).

2 Theory: what a pair of flip angles buys

Write $c(x) = x/\sin x$ (x in radians) and $h = \tau/(2\sinh\tau)$ with $\tau = T_R/T_1$; $h \approx 1/2$ for $T_1 \gg T_R$.

Proposition 1 (Decomposition)**.** *The Fisher information of a two-scan design at transmit scale B_1^+ is*

$$\mathcal{I}(\alpha_1, \alpha_2; B_1^+) = \mathcal{I}_1(B_1^+\alpha_1) + \mathcal{I}_1(B_1^+\alpha_2), \tag{2}$$

where $\mathcal{I}_1(x)$ is the information of a single scan at actual flip angle x, and the $\ln B_1^+$ column of a single-scan Jacobian is $\partial/\partial \ln x$.

Proof. The scans have independent noise, so their informations add. Within one scan B_1^+ enters only through $x = B_1^+\alpha$, so $\partial/\partial \ln B_1^+ = x\,\partial/\partial x = \partial/\partial \ln x$. □

Proposition 1 makes an exhaustive search cheap: $\mathcal{I}_1$ is tabulated once per tissue on the actual angles that occur, and every design at every B_1^+ is a sum of two 7×7 matrices. It agrees with direct evaluation of the two-scan information to 3×10^{-15}.

Proposition 2 (One scan in Ernst-angle coordinates)**.** *Fix T_2, T_2^*, $\Delta\omega$, ϕ_0. The signal of one scan depends on (M_0, B_1^+, T_1) only through the offset w of (1) and $\ell = \ln(M_0\zeta)$, with*

$$\frac{\partial w}{\partial \ln B_1^+} = c(x), \qquad \frac{\partial w}{\partial \ln T_1} = h, \qquad \frac{\partial \ell}{\partial \ln T_1} = -h, \qquad \frac{\partial \ell}{\partial \ln M_0} = 1. \tag{3}$$

Consequently its Jacobian columns satisfy $J_{T_1} = h\,(J_{B_1^+}/c - J_{M_0})$, and its information about (w, ℓ) depends on the flip angle only through w (for given ℓ, T_2, T_2^*, $\Delta\omega$). Angles with equal $|\tan(x/2)|$, i.e. x and $\pm x + 360°m$, therefore carry the same information and differ only in the rate c: $\mathcal{I}_1(x') = D\,\mathcal{I}_1(x)\,D$, where D multiplies the $\ln B_1^+$ row and column by $c(x')/c(x)$.

Proof. By the two-invariant form and the Ernst-angle form,

$$M_0 a_k = \operatorname{sign}\!\big(\tan\tfrac{x}{2}\big)\,\mathrm{e}^{\ell}\operatorname{sech}(w)\,G_k(\tanh w, E_2, \Delta\omega),$$

which depends on (M_0, B_1^+, T_1) through (w, ℓ) only; the sign is constant between multiples of 180° (it changes sign there, which is what produces aliases, Section 4) and does not enter the information. The derivatives follow from $\zeta^2 = \tanh(\tau/2)$ and

$$\frac{\mathrm{d}\ln|\tan(x/2)|}{\mathrm{d}\ln x} = \frac{x}{\sin x} = c, \qquad \frac{\mathrm{d}\ln\zeta}{\mathrm{d}\ln\tau} = \frac{\tau}{2\sinh\tau} = h, \qquad \frac{\mathrm{d}\ln\tau}{\mathrm{d}\ln T_1} = -1.$$

The information about (w, ℓ) is Re of the Gram matrix of $(\partial S/\partial w, \partial S/\partial \ell)$, a function of w alone. □

The identity $J_{T_1} = h(J_{B_1^+}/c - J_{M_0})$ holds numerically to 10^{-14} at the angles tested. It is the precise sense in which a single scan cannot separate B_1^+ from T_1: the T_1 column is a fixed combination of the other two. It also says that a 330° pulse is a 30° pulse ($|\tan 165°| = \tan 15°$) whose sample moves eleven times faster with B_1^+, and in the opposite direction (we call $c(x)$ the *rate* of a scan): $c(30°) = 1.05$, $c(330°) = -11.5$ (Fig. 1a,b). The per-scan information on w (Fig. 1c) peaks a little above the Ernst angle for the parenchymal tissues ($w = 0.1$–0.4, actual angles 13–16°) and well above it for CSF ($w = 1.85$, 39°), whose long T_2 makes the pathway ratios informative at larger angles.

Proposition 3 (Two scans: closed-form bounds)**.** *Let $L = c(x_2) - c(x_1) \neq 0$ be the* lever *of the design (the difference of the per-scan rates c), and let Q be the information about (w_1, w_2) after the shared amplitude ℓ and the parameters $\ln T_2$, $\ln T_2^*$, $\Delta\omega$, ϕ_0 have been profiled out. Then the Cramér–Rao bounds of the full model are*

$$\mathrm{SD}(\ln B_1^+) = \frac{\sqrt{e^{\mathsf{T}}Q^{-1}e}}{|L|}, \quad e = (-1, 1), \qquad \mathrm{SD}(\ln T_1) = \frac{\sqrt{g^{\mathsf{T}}Q^{-1}g}}{h\,|L|}, \quad g = (c_2, -c_1), \tag{4}$$

and with B_1^+ known $\mathrm{SD}(\ln T_1) = 1/(h\sqrt{\mathbf{1}^{\mathsf{T}}Q\mathbf{1}})$. With T_2, T_2^*, $\Delta\omega$, ϕ_0 known, the same formulas hold with Q profiled over ℓ only.

Proof. By (3), the map $\theta \mapsto (w_1, w_2, \ell, \ln T_2, \ln T_2^*, \Delta\omega, \phi_0)$ is a local diffeomorphism whose Jacobian has the block $\left(\begin{smallmatrix} 0 & c_1 & h \\ 0 & c_2 & h \\ 1 & 0 & -h \end{smallmatrix}\right)$ in $(\ln M_0, \ln B_1^+, \ln T_1)$, of determinant $-hL$, and the identity in the remaining parameters. Inverting it, to first order $\mathrm{d}\ln B_1^+ = (\mathrm{d}w_2 - \mathrm{d}w_1)/L$ and $h\,\mathrm{d}\ln T_1 = (c_2\,\mathrm{d}w_1 - c_1\,\mathrm{d}w_2)/L$. The bounds transform with the Jacobian, so the covariance of $(\ln B_1^+, \ln T_1)$ is that of (w_1, w_2), i.e. Q^{-1} (the inverse of the information profiled over all other coordinates), mapped by these two linear forms. With B_1^+ known, $\mathrm{d}w_1 = \mathrm{d}w_2 = h\,\mathrm{d}\ln T_1$, so the information about $h\ln T_1$ is $\mathbf{1}^\mathsf{T} Q \mathbf{1}$. □

Proposition 3 separates the design into two ingredients: *where* each scan samples the bump (through Q, which depends on the offsets w_1, w_2) and *how differently* the two samples move with B_1^+ (through L). B_1^+ is measured by the difference of the offsets divided by the lever, T_1 by a lever-weighted combination. Writing $g = L\,e_1 - c_1 e$ with $e_1 = (1, 0)$ gives the exact decomposition

$$\mathrm{Var}(\ln T_1) = \frac{(Q^{-1})_{11} + 2(c_1/L)\big[(Q^{-1})_{11} - (Q^{-1})_{12}\big]}{h^2} + \Big(\frac{c_1}{h}\Big)^2 \mathrm{Var}(\ln B_1^+)\,: \tag{5}$$

the T_1 uncertainty is a scan-1 term plus the B_1^+ uncertainty carried over with the factor $c_1/h \approx 2c_1$, the same factor that carries a pulse-ratio error (Proposition 4). Three consequences follow.

- *Large levers.* As $|L|$ grows, $\mathrm{Var}(\ln T_1) \to (Q^{-1})_{11}/h^2 + (c_1/h)^2 \mathrm{Var}(\ln B_1^+)$, with a correction of order c_1/L. For the published design ($L = -12.5$) the correction lowers the variance by about a fifth (WM: from 937 to 746); for $|L| \approx 6$, as in the no-crossing designs, it is comparable to the total, and the limit is not a useful approximation. The lever is largest where $\sin x_2 \to 0$, i.e. at $x_2 = 180°m$, but there the scan-2 signal vanishes too. At such a crossing Proposition 3 holds only as a limit; directly, the scan-2 Jacobian then has a single non-zero column, $\ln B_1^+$ (the signal is zero but its derivative is not), so scan 2 measures B_1^+ alone, a null measurement. Because the single-scan null direction $(\delta\ln M_0, \delta\ln B_1^+, \delta\ln T_1) \propto (h, -h/c_1, 1)$ has a non-zero B_1^+ component, the information matrix stays non-singular (the bounds finite) unless x_1 is also at a crossing.

- *Small angles.* For two small angles $c_1 \approx c_2 \approx 1$, $L \to 0$, and both bounds diverge: the leading-order degeneracy of small-angle protocols [2].

- *Equivalent pulses.* 30°, 330°, 390° and 690° have the same per-scan information but rates $c = 1.05$, -11.5, 13.6 and -24.1. A larger rate helps only if the pulse does not cross a multiple of 180° over the range of B_1^+ and is affordable in SAR.

Proposition 4 (Pulse-ratio error)**.** *Suppose the actual scan-2 angle is $(1+\varepsilon)$ times its nominal value relative to scan 1 and the fit assumes the nominal ratio. Then, to first order and whichever of T_2, T_2^*, $\Delta\omega$, ϕ_0 are fitted,*

$$\delta\ln T_1 = -\frac{c_1 c_2}{h\,L}\,\varepsilon, \qquad \delta\ln B_1^+ = \frac{c_2}{L}\,\varepsilon, \qquad \delta\ln M_0 = h\,\delta\ln T_1, \tag{6}$$

T_2, T_2^*, $\Delta\omega$, ϕ_0 are unbiased, and the fit has zero residual: the error cannot be detected from the data.

Proof. The error shifts the scan-2 offset by $c_2\varepsilon$ and nothing else, so the perturbation of the data is $\delta S = \varepsilon c_2\,\partial S/\partial w_2$. By the proof of Proposition 3 this equals $J\,\delta\theta^*$ with $\delta\theta^*$ given by (6) (and $\delta\ell = 0$, hence $\delta\ln M_0 = h\,\delta\ln T_1$) and zero in the other parameters. Since δS lies in the range of J,

the first-order least-squares bias $\mathcal{I}^{-1}J^{\mathsf{T}}\delta S = \mathcal{I}^{-1}J^{\mathsf{T}}J\,\delta\theta^* = \delta\theta^*$, for any set of free parameters with J of full column rank. For finite ε, the local invertibility of the map of Proposition 3 (for $L \neq 0$, and as long as $(1+\varepsilon)x_2$ crosses no multiple of 180°) gives parameters that reproduce the perturbed data exactly. □

The transfer factor can be written $-c_1c_2/(hL) = -1/\big(h\,[\operatorname{sinc} x_1 - \operatorname{sinc} x_2]\big)$ with $\operatorname{sinc} x = \sin x/x$. Since $|\operatorname{sinc} x| \le 1$ and $\operatorname{sinc} x \ge -0.2172$ (the minimum, at $x = 257.45°$ where $\tan x = x$), every design has $|\mathrm{d}\ln T_1/\mathrm{d}\varepsilon| \ge 1/(1.2172\,h) \approx 1.64$, approached for $x_1 \to 0$ and $x_2 \to 257°$. No choice of angles makes the factor small: a 1% ratio error becomes at least a 1.6% T_1 error (Section 6).

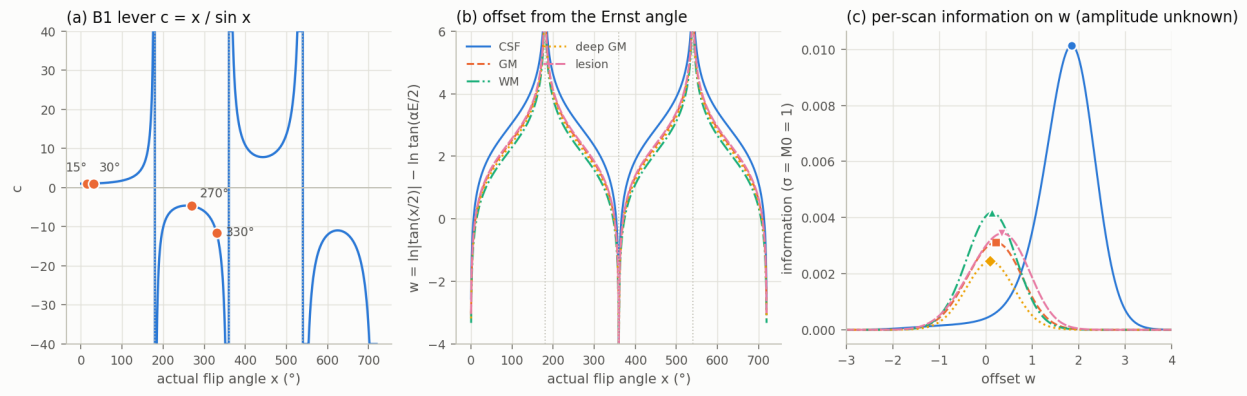


Figure 1: (a) The rate $c(x) = x/\sin x$ at which a sample moves along the log half-angle axis per unit change of $\ln B_1^+$. Dotted: multiples of 180°. (b) The offset (1) of a sample from each tissue's Ernst angle; x and $360° - x$ have the same offset. (c) Information about w in one scan with the amplitude unknown (T_2, T_2^*, $\Delta\omega$, ϕ_0 known), against w; markers: maxima.

Table 2 evaluates Propositions 3 and 4 for the shortlisted designs of Section 5; the closed forms agree with direct inversion of the 3×3 and 7×7 information to rounding. Freeing T_2, T_2^*, $\Delta\omega$ and ϕ_0 costs 7–30% in the T_1 bound; in (4) this is the information about the offsets that scan 2 must spend on T_2 and T_2^*.

Table 2: Propositions 3 and 4 for the shortlist, WM at $B_1^+ = 1$ (bounds per unit σ/M_0). "known": T_2, T_2^*, $\Delta\omega$, ϕ_0 known; "free": all seven parameters free. Ratio bias: $\mathrm{d}\ln T_1/\mathrm{d}\varepsilon$ and $\mathrm{d}\ln M_0/\mathrm{d}\varepsilon$. Last column: largest relative difference between the closed forms and direct inversion of the 3×3 and 7×7 information. For 28°/180°, $x_2 = 180°$ at $B_1^+ = 1$ and the closed form is a limit.

		SD($\ln B_1^+$)		SD($\ln T_1$)		ratio bias		closed vs
α_1/α_2	L	known	free	known	free	T_1	M_0	matrix
15/330	−12.53	2.16	2.21	25.5	27.3	−1.86	−0.93	1×10^{-15}
20/276	−5.86	3.81	6.23	26.7	32.7	−1.69	−0.84	1×10^{-15}
14/309	−7.95	3.27	5.09	25.7	30.3	−1.76	−0.88	4×10^{-16}
21/270	−5.74	3.94	6.25	27.5	33.4	−1.68	−0.84	2×10^{-15}
17/254	−5.63	4.47	8.33	25.2	32.6	−1.66	−0.83	1×10^{-15}
28/180	∞	7.00	7.00	44.6	53.0	−2.08	−1.04	(limit)

3 The local design maps

Figure 2 shows the worst-case and mean T_1 bounds and the worst-case B_1^+ bound over all pairs with $\alpha_1 \leq 90°$. Pairs with $\alpha_1 > 90°$ are never competitive: their best worst-case T_1 bound is 1499, fifteen times the optimum. Designs with a complex alias in the fitting range (Section 4) are greyed out; contours mark 1.05 and 1.1 times the best value among the remaining designs.

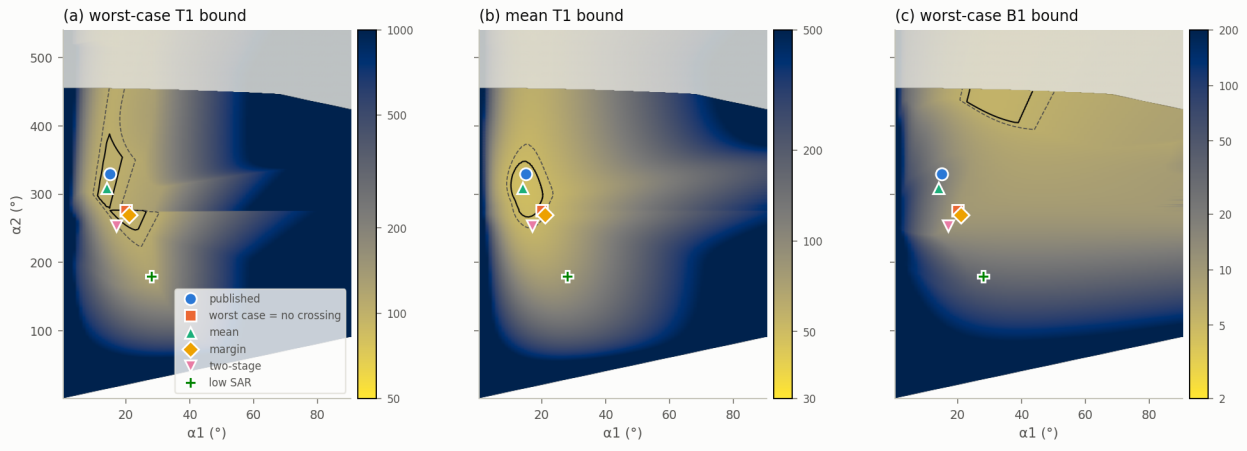


Figure 2: Design maps over the nominal angles (1° grid; bounds per unit σ/M_0, all parameters free). (a) Worst case of the T_1 bound over $B_1^+ \in [0.7, 1.3]$ and the five tissues; (b) its mean; (c) worst case of the B_1^+ bound. Grey: a complex alias in the fitting range $[0.6, 1.4]$. Solid and dashed contours: 1.05 and 1.1 times the best value among designs without a complex alias. Markers: the shortlist of Table 4.

The optimum is flat, and the published design is in it. The best mean T_1 bound, 49.2, is reached at 14°/309° (flat to 0.2% for $\alpha_2 = 305$–$312°$; the mean is an average over the 0.01 grid of B_1^+); the published 15°/330° gives 50.2, 2% more. The designs within 5% of the optimum form a curved band (solid contour in Fig. 2b) that extends over $\alpha_1 = 11$–$20°$ and $\alpha_2 = 267$–$348°$: at $\alpha_1 = 15°$ every α_2 from 267° to 348° qualifies, at 11° only 299–329° and at 20° only 291–309°. The best worst-case bound, 100.3, is reached at 20°/276°; the published design gives 103.6, 3.3% more. In terms of T_1 precision alone, the flip angles of the published protocol cannot be improved materially. The best α_1 follows from Fig. 1c: the information about the offset peaks at actual angles of 13–16° for the parenchymal tissues, just above their Ernst angles, and the lever of a near-360° pulse does the rest.

The worst case is decided by a narrow band. The worst case of every near-360° design is set by CSF where $B_1^+ \alpha_2 = 360°$ (below). Outside a band of ±0.5% in B_1^+ around that crossing, the published design's worst case is 91.9 and the best design (12°/332°) reaches 89; 20°/276° wins only because its crossing lies just outside the range, at $B_1^+ = 1.304$. Its α_2 is therefore set by the assumed upper limit of B_1^+ ($\alpha_2 = \lfloor 360°/1.3 \rfloor$): the worst case jumps from 100.3 at $\alpha_2 = 276°$ to 113.4 at 277°. For parenchyma alone (no CSF) the best worst case is 61.0 at 15°/308°, against 68.0 for the published design (−10%).

What the angles do decide. Within the flat region the designs differ in the secondary quantities. Figure 3 shows the bounds against B_1^+ for the shortlist.

- *The* 360° *crossing.* For α_2 between 277° and 514° the scan-2 pulse passes through 360° somewhere in $B_1^+ \in [0.7, 1.3]$ (at $B_1^+ = 1.09$ for 330°). There the scan-2 signal vanishes and scan 2 measures only B_1^+ (Proposition 3): the B_1^+ bound has a deep minimum, and the T_1 bound rises to that of scan 1 alone with B_1^+ known. The rise comes from the loss of scan 2's information about T_2 and T_2^*; with these known there is no peak, and at 180° crossings there is none either. The peak is about 0.5% wide in B_1^+ for CSF and 1–1.3% for the parenchymal tissues, and it sets the worst case of the published design (CSF, 103.6 against a mean of 82.8).

- *Equal offsets.* In tissues with a short T_2 the T_1 bound also rises where the two scans sample the same offset, $w_1 = w_2$, i.e. $x_2 = 360° \pm x_1$ ($B_1^+ = 1.04$ and 1.14 for 15°/330°): there the pathway ratios of the two scans coincide and separate T_2 from T_1 less well (Fig. 3a).

- *No crossing.* 20°/276° keeps both pulses away from multiples of 180° over the whole range, with no headroom at the top (359° at $B_1^+ = 1.3$, where its bound rises). Its worst-case B_1^+ bound is 10.2 against 19.2, and its scan-2 pulse needs 30% less energy at equal pulse duration (16% less, with a 16% shorter pulse, at equal peak B_1^+). It pays 5% in the mean T_1 bound, mostly in CSF (91.3 against 82.8); only WM is slightly better (36.8 against 37.4), while GM, deep GM and lesion lose 2–4%. The design with 2.5% headroom at both ends, 21°/270° (B_1^+ crossings at 0.667 and 1.333), has nearly the same worst case (101.4) and B_1^+ bound (10.2); 18°/270° trades 1% of worst case for a better mean (51.8 against 53.7) and a scan-1 T_1 bound within 0.2% of the two-stage optimum (109.9 against 109.7 at $\alpha_1 = 17°$).

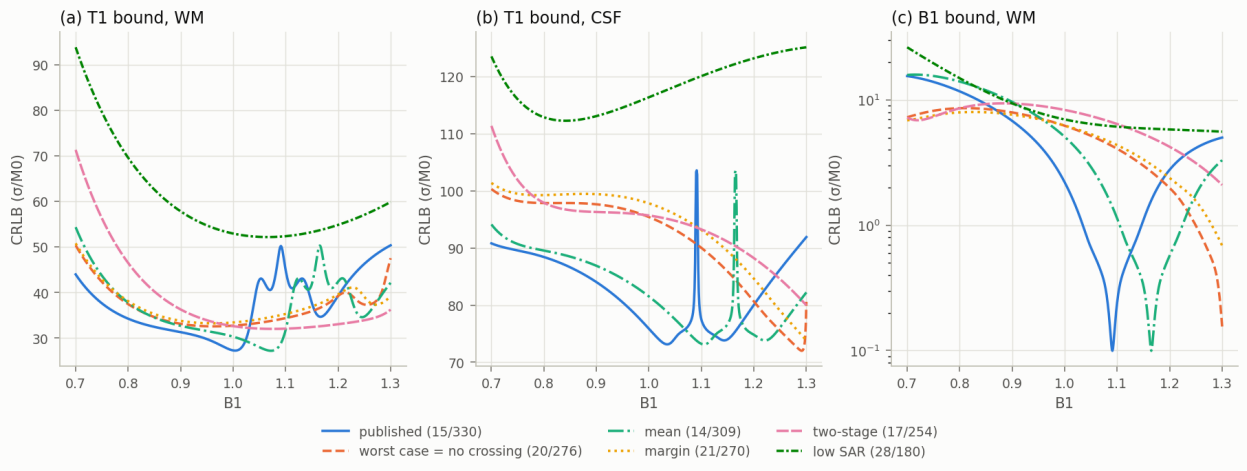


Figure 3: Bounds against B_1^+ for the shortlisted designs (all parameters free; the B_1^+ grid includes each design's crossing points). (a), (b) T_1 bound for WM and CSF: the near-360° designs show a narrow peak where $B_1^+ \alpha_2 = 360°$, and in WM broader bumps where $w_1 = w_2$. (c) B_1^+ bound for WM, logarithmic: the same crossing gives a null measurement of B_1^+.

4 Global identifiability: aliases

The bounds of Section 3 are local. A design can have small bounds and still be unusable if a different set of parameters reproduces the data exactly.

Definition 1 (Alias)**.** For true parameters $\theta = (M_0, B_1^+, T_1, T_2, T_2^*, \Delta\omega, \phi_0)$, an *alias* is a parameter vector $\theta' = (M_0', B_1^{+\prime}, T_1', T_2, T_2^*, \Delta\omega, \phi_0')$ with $B_1^{+\prime} \neq B_1^+$ and a physical T_1' (an Ernst angle in $(0°, 90°)$, i.e. $0 < \tan(\alpha_E'/2) < 1$) whose noise-free signals equal those of θ in every pathway, echo and scan. It is a *complex alias* if the complex signals agree (with $\phi_0' = \phi_0 + \pi$ allowed, which absorbs a sign change of both scans), and a *magnitude alias* if only their magnitudes do.

T_2, T_2^* and $\Delta\omega$ are held at their true values because the echo decay and the echo phases determine them (up to phase wraps of $\Delta\omega$); the characterisation below also uses that the pattern of pathway amplitudes determines u, which holds numerically for the tissues considered.

An alias is not a nearby value and not a poorly determined one: the least-squares cost has an equally deep minimum at θ', and for noisy data every minimum near the truth has a twin near θ' with identical cost. No signal-to-noise ratio separates them; only prior knowledge (a fitting range for B_1^+) or additional data can. This distinguishes an alias from a local minimum (higher cost, avoidable with better starting values) and from poor conditioning (a unique but noisy solution).

Where aliases come from. Two scans determine w_1, w_2 and ℓ; complex data with a phase shared by the two scans also determine the relative sign of v_1 and v_2. Dividing the two half-angle ratios $\xi_i = \tan(x_i/2)/\zeta$ eliminates T_1 and M_0 and leaves one equation in B_1^+ alone,

$$f(B_1^+) = \frac{\tan(B_1^+\alpha_2/2)}{\tan(B_1^+\alpha_1/2)} = \frac{\xi_2}{\xi_1}, \tag{7}$$

whose right side the data fix (Fig. 4). Every other root $B_1^{+\prime}$ of (7) with

$$\tan(\alpha_E'/2) = \zeta \left| \frac{\tan(B_1^{+\prime}\alpha_1/2)}{\tan(B_1^+\alpha_1/2)} \right| < 1$$

is a complex alias, with $M_0' = M_0\zeta/\tan(\alpha_E'/2)$; every admissible root of $f(B_1^{+\prime}) = -f(B_1^+)$ is a magnitude alias, in which the scan-2 signal is negated relative to scan 1. The alias values of B_1^+ depend on the true B_1^+ alone, not on the tissue; the tissue enters only through T_1' and through admissibility. For the published design and true $B_1^+ = 1$ ($f = -2.04$) the magnitude alias is $B_1^{+\prime} = 1.199$ (the scan-2 pulse becomes $395.7° \equiv +35.7°$ instead of $330° \equiv -30°$; $T_1' = 588\,\text{ms}$ for WM) and the complex alias is $B_1^{+\prime} = 2.008$ ($T_1' = 203\,\text{ms}$), excluded by any fitting range below $B_1^+ = 540/330$; more generally, a fitting range inside $(180°/\alpha_2, 540°/\alpha_2) = (0.545, 1.636)$ excludes every complex alias for true B_1^+ in it, because f is monotonic on that branch.

Local and global identifiability are linked. Differentiating (7),

$$\frac{\mathrm{d}\ln|f|}{\mathrm{d}\ln B_1^+} = c(B_1^+\alpha_2) - c(B_1^+\alpha_1) = L, \tag{8}$$

the lever of Proposition 3. Since $f' = f\,L/B_1^+$, f is monotonic on an interval between consecutive poles ($x_2 = 180° + 360°m$ or $x_1 = 360°m$) exactly where $f\,L$ keeps its sign; L itself changes sign only at the zeros of f, where f' stays finite. A complex alias on the same branch therefore requires a point with $L = 0$, where the information is singular. Magnitude aliases arise around each zero of f ($x_2 = 360°m$ or $x_1 = 180° + 360°m$) and each pole ($x_2 = 180° + 360°m$) inside the fitting range, where f changes sign.

Proposition 5 (No aliases without crossings)**.** *If over the whole fitting range $x_1 = B_1^+\alpha_1 \in (0°, 180°)$ and $x_2 = B_1^+\alpha_2 \in (180°, 360°)$, the design has no alias of either kind in that range.*

Proof. There $\tan(x_1/2) > 0 > \tan(x_2/2)$, so $f < 0$, and $c_1 > 0 > c_2$, so $L < 0$; by (8) $|f|$ is strictly decreasing, hence one-to-one, and $f(B_1^{+\prime}) = \pm f(B_1^+)$ has no root other than B_1^+. □

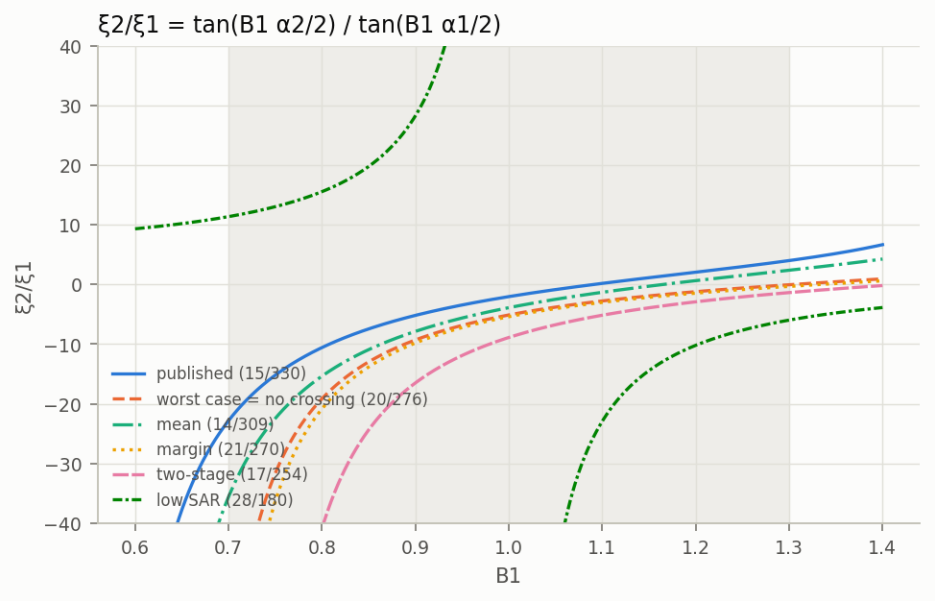


Figure 4: The B_1^+-only quantity (7) for the shortlist over the fitting range $[0.6, 1.4]$ (shaded: robustness range). A complex alias exists if a horizontal line meets a curve twice within the fitting range; a magnitude alias if it meets the curve and its mirror image $-f$.

Aliases over the design space. With the fitting range $[0.6, 1.4]$, of the 145 530 pairs 94 540 have a complex alias for some true B_1^+ in $[0.7, 1.3]$ (mostly pairs with a large α_1 or an α_2 above about 450°, greyed in Fig. 2), 42 293 have only magnitude aliases and 8 697 have none. This fitting range spans a factor 2.3 in B_1^+, more than any pulse can stay within $(180°, 360°)$, so every near-optimal design has a magnitude alias somewhere in it: the best design without any alias has a worst-case T_1 bound of 197, twice the optimum. With $[0.6, 1.4]$ a good protocol therefore needs complex data, or for magnitudes the one bit of phase supplied by the FID-sign test [2]. A narrower fitting range changes this for the no-crossing designs: fitted within their own crossing-free range (e.g. $B_1^+ \in [0.667, 1.333]$ for $\alpha_2 = 270°$, which still contains the robustness range), they have no alias at all (Proposition 5), and magnitude-only fitting becomes possible. All shortlisted designs are free of complex aliases on $[0.6, 1.4]$; Table 4 lists where their magnitude aliases occur. For the published design they cover most of the range (true $B_1^+ \geq 0.87$); for the no-crossing designs only its ends, near the crossings that lie just outside it, with the alias 1–16% from the truth and near the edge of the fitting range.

5 Constraints and shortlist

SAR and peak B_1^+. Figure 5a gives the best joint design (no complex alias) as a function of the largest nominal angle. Both curves are flat above about 280° (worst case) and 310° (mean): larger pulses buy nothing. Below, the price rises steadily. A limit of 240° costs 6% in the worst case and 14% in the mean; a limit of 180° costs 25% and 52%; a limit of 120° more than doubles both. Relative to 330°, a second pulse at 276° needs 30% less energy at equal pulse duration; at equal peak B_1^+ it is 16% shorter and needs 16% less energy.

A second pulse below 90°. A small second pulse is attractive for SAR, but it removes the lever. Both pulses then lie in the regime where $c(x) = x/\sin x$ barely changes, so L is small (+0.48 for 15°/85°, against −12.5 for 15°/330°) and the two scans see B_1^+ and T_1 almost as one scan does. Table 3 gives the price. At 85° the worst-case joint T_1 bound is 6.8 times the published one with $\alpha_1 = 15°$ and still 4.5 times with the best first angle (34°); the worst-case B_1^+ bound is 10–17 times larger. The loss comes from having to estimate B_1^+: with B_1^+ known and $\alpha_1 = 15°$, the scan-pair T_1 bound is 93–107 for every α_2 in the table. In exchange, scan 2 needs 7% of the energy of 330°, there is no alias and no 360° crossing, and finite-pulse errors are small. The cost falls steadily as α_2 rises and is moderate only beyond 180°, where c changes sign and $|L|$ becomes large: 210° costs 10–60% at 40% of the energy, and 270° costs 0–10% at 67%.

Table 3: Cost of a smaller second pulse. For each α_2: $\alpha_1 = 15°$ and the α_1 with the smallest worst-case joint T_1 bound (no complex alias). Lever $L = c(\alpha_2) - c(\alpha_1)$ at $B_1^+ = 1$; bounds per unit σ/M_0, worst case (w.) or mean over $B_1^+ \in [0.7, 1.3]$ and the five tissues; "B_1^+ known": joint two-scan T_1 bound with B_1^+ fixed; "rel.": worst-case T_1 bound relative to 15°/330°; energy: scan-2 RF energy per pulse relative to 330° at equal pulse duration.

		T_1 bound		B_1^+	T_1, B_1^+ known		
α_1/α_2	L	w.	mean	w.	w.	rel.	energy
15/85	+0.48	701	280	322	99	6.8	0.07
34/85	+0.43	461	229	197	133	4.5	0.07
15/120	+1.41	396	160	166	103	3.8	0.13
33/120	+1.36	238	137	88	164	2.3	0.13
15/150	+4.22	267	112	99	104	2.6	0.21
31/150	+4.19	160	103	51	150	1.5	0.21
15/210	−8.34	164	67	37	107	1.6	0.40
25/210	−8.36	114	68	20	113	1.1	0.40
15/270	−5.72	109	51	15	93	1.1	0.67
21/270	−5.74	101	54	10	99	1.0	0.67
15/330	−12.53	104	50	19	104	1.0	1.00

Two-stage pipeline. When scan 2 serves only to map a smooth B_1^+, α_1 should minimise the scan-1 T_1 bound with B_1^+ known (Fig. 5b). The minimum lies at 17° (worst case, 109.7) to 19° (mean, 64.5), against 117.8 and 68.0 at 15°: a gain of 5–7%, consistent with the information peak just above the Ernst angles (Fig. 1c). Between 15° and 22° the worst case varies by about 7%. Because smoothing makes the noise in the B_1^+ map negligible [2], α_2 is chosen for an alias-free, well-conditioned B_1^+ measurement and for robustness; restricted to $\alpha_2 \leq 330°$ (no more SAR than the published design), the smallest worst-case B_1^+ bound for $\alpha_1 = 17°$ is at $\alpha_2 = 254°$ (without the restriction it lies near 440°, a larger lever at 78% more energy, Fig. 2c). This pulse crosses 180° at $B_1^+ = 0.71$, which costs nothing in the T_1 bound but produces magnitude aliases for B_1^+ below 0.8; $\alpha_2 = 258$–270° avoids the crossing (e.g. 17°/270°: worst-case B_1^+ bound 12.3 against 11.2).

Shortlist. Table 4 collects the designs carried forward. The rules that select them are: the published design; the minimum worst-case and minimum mean joint T_1 bound; the best design whose pulses cross no multiple of 180° for $B_1^+ \in [0.7, 1.3]$ (it coincides with the worst-case optimum, $\alpha_2 = 258$–276°); the best such design with 2.5% headroom at both ends of the range (*margin*, $\alpha_2 = 264$–270°); the two-stage design above; and the best design with both angles at most 180°.

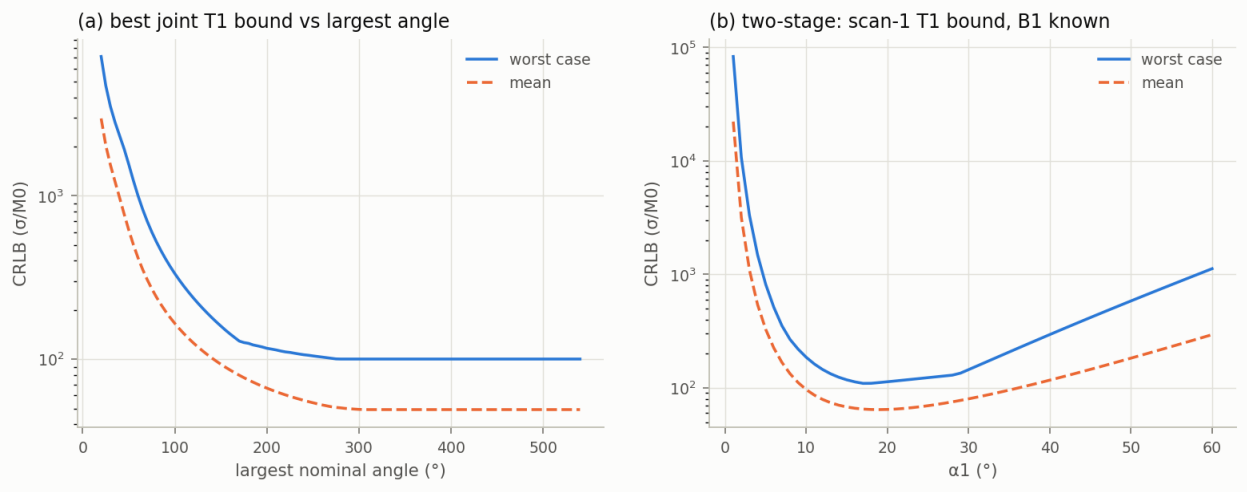


Figure 5: (a) Best worst-case and mean joint T_1 bound among designs without a complex alias whose largest angle does not exceed the abscissa. (b) Two-stage pipeline: scan-1 T_1 bound with B_1^+ known against α_1 (worst case and mean over $B_1^+ \in [0.7, 1.3]$ and the tissues).

All are free of complex aliases in the fitting range; the last column gives the true B_1^+ for which a magnitude alias exists.

Table 4: Shortlisted designs. Bounds per unit σ/M_0, all parameters free, worst case (w.) or mean over $B_1^+ \in [0.7, 1.3]$ and the five tissues; "par.": parenchyma only (no CSF). Scan-1 T_1: two-stage criterion (B_1^+ known). Ratio bias: largest $|\mathrm{d} \ln T_1 / \mathrm{d}\varepsilon|$ over the range. Last column: true B_1^+ with a magnitude alias in the fitting range $[0.6, 1.4]$ (no design has a complex alias).

		T_1 bound			B_1^+	T_2	scan-1 T_1	ratio	
Design	α_1/α_2	w.	mean	w. par.	w.	w.	w., B_1^+ known	bias	magn. alias
published	15/330	103.6	50.2	68.0	19.2	117.3	117.8	2.34	0.87–1.30
worst case = no crossing	20/276	100.3	52.6	73.9	10.2	115.9	113.4	2.06	1.22–1.30
mean	14/309	103.6	49.2	61.9	20.5	130.1	124.3	2.25	0.99–1.30
margin	21/270	101.4	53.7	63.7	10.2	114.8	115.3	2.02	0.70–0.72, 1.27–1.30
two-stage	17/254	111.4	53.6	78.5	11.2	124.0	109.7	2.04	0.70–0.79
low SAR	28/180	125.1	81.9	117.0	32.0	127.6	129.7	3.26	0.70–1.00, 1.00–1.21

6 Robustness to model error

The bounds assume that the fitting model is exact. Table 5 generates noise-free data with an effect the ideal model ignores and fits them with the ideal two-scan model (complex, all seven parameters free, ten starting points, B_1^+ restricted to the fitting range $[0.6, 1.4]$) for WM, GM and CSF at $B_1^+ = 0.7, 0.8, \ldots, 1.3$ and at the transmit scales where the design's scan-2 angle is 330° and 345°.

- *Pulse ratio*: the scan-2 angle 1% larger than nominal relative to scan 1, as produced by an imperfect calibration of two pulses of different shape or duration.
- *Slab-edge ramp*: the flip angle falls linearly from 100% to 90% across the voxel, each position in its own steady state, as at the edge of a 3D slab.

- *Finite pulses*: hard pulses at a fixed peak B_1^+, so that the duration is proportional to the nominal angle (1 ms for 330°), with relaxation during the pulse; echo times are measured from the pulse centre. On resonance, and with 50 Hz off resonance (precession during the pulse).

Besides the bias we give the probability that the misfit would be flagged at SNR 300 by a χ^2 test on the residual (level 99.9%, 29 degrees of freedom; noncentral χ^2 with the noise-free residual as noncentrality).

Table 5: Bias of T_1, $100(\hat{T}_1/T_1 - 1)$ (%), when the ideal two-scan model is fitted to data with the effect (noise-free, all parameters free). Columns 3–5: at $B_1^+ = 1$; column 6: largest |bias| over WM, GM, CSF and the sampled B_1^+ ($0.7, 0.8, \ldots, 1.3$ and the scan-2 angles 330°, 345°), a lower bound for the continuous range; column 7: largest $|\delta \ln T_1|$ divided by the T_1 standard deviation at SNR(M_0) = 300 (bound/300); column 8: probability that the misfit of the largest-bias case is flagged at SNR 300. *: a fit ended on a safeguard bound of the optimiser. Generated by `scripts/flip_design_robustness.py`.

		$B_1^+ = 1$			largest over the sampled cases		
α_1/α_2	effect	WM	GM	CSF	|bias| (%)	|bias|/SD	flagged
15/330	pulse ratio +1%	−1.8	−1.8	−1.8	2.3	0.2	0.00
	slab-edge ramp	+1.7	+0.9	+0.7	40.3	1.3	0.58
	finite pulses	+19.7	+18.5	+5.4	19.7	2.0	0.04
	finite pulses, 50 Hz	+22.2	+19.6	+11.9	22.2	2.2	0.99
20/276	pulse ratio +1%	−1.7	−1.7	−1.7	2.1	0.2	0.00
	slab-edge ramp	−0.3	−0.3	−0.2	16.3	1.1	0.10
	finite pulses	+2.4	+2.2	+0.6	13.8	1.2	0.01
	finite pulses, 50 Hz	+2.7	+2.5	+3.8	13.3	1.1	0.92
14/309	pulse ratio +1%	−1.8	−1.8	−1.8	2.2	0.2	0.00
	slab-edge ramp*	−0.1	−0.3	0.0	74.1	3.9	0.83
	finite pulses	+6.4	+5.1	+0.6	19.5	2.0	0.03
	finite pulses, 50 Hz	+6.9	+5.6	+8.4	21.9	2.2	0.98
21/270	pulse ratio +1%	−1.7	−1.7	−1.7	2.0	0.2	0.00
	slab-edge ramp	−0.3	−0.3	−0.1	13.0	0.8	0.02
	finite pulses	+2.2	+2.1	+1.8	14.4	1.2	0.01
	finite pulses, 50 Hz	+2.5	+2.4	+3.5	13.9	1.2	0.88
17/254	pulse ratio +1%	−1.6	−1.6	−1.6	2.0	0.2	0.00
	slab-edge ramp	−0.2	−0.2	0.0	0.9	0.1	0.00
	finite pulses	+2.0	+1.9	+2.1	10.0	0.8	0.05
	finite pulses, 50 Hz	+2.3	+2.2	+3.1	11.1	0.9	0.88
28/180	pulse ratio +1%	−2.0	−2.0	−2.0	3.1	0.1	0.00
	slab-edge ramp	−0.2	−0.2	0.0	0.4	0.0	0.00
	finite pulses	+2.0	+2.0	+1.8	3.4	0.1	0.00
	finite pulses, 50 Hz	+2.3	+2.3	+2.2	4.0	0.1	0.00

Pulse ratio: the same for every design, and invisible. A 1% ratio error biases T_1 by −1.6 to −2.0% at $B_1^+ = 1$ in every design (up to −3.1% over the B_1^+ range), as Proposition 4 predicts, and the fit absorbs it with zero residual. No choice of angles avoids it ($|\mathrm{d} \ln T_1/\mathrm{d}\varepsilon| \geq 1.64$): the ratio must be calibrated, to about 0.5% if T_1 is to be accurate to 1%. The bias is only 0.2 standard deviations per voxel at SNR 300, but it is systematic and does not average out over a region.

Everything goes wrong near 360°**.** The other two effects share one pattern: they bias T_1 strongly where the actual scan-2 angle approaches 360°, i.e. where its effective angle is small (finite pulses most at 330–350°, flip-angle spreads at and across 360°), and hardly at all elsewhere; crossings of 180° are harmless, because the scan-2 signal there is small in any case ($|v| \approx \zeta^2|\delta|$ at $180° + \delta$ against $|\delta|$ at $360° + \delta$).

- *Finite pulses.* With a 1 ms 330° pulse the published design overestimates T_1 by 20% in WM and 18.5% in GM at $B_1^+ = 1$, two standard deviations at SNR 300, with a misfit that is flagged in only 2–4% of voxels. The margin design 21°/270° gives 2.2% and 2.1% there. The bias grows with the pulse duration: for a 330° pulse of 0.125, 0.25, 0.5, 1, 2 and 4 ms it is 3.4, 6.6, 12.7, 19.7, 22.6 and 33.6% for the published design (steep, then irregular) and 0.27, 0.54, 1.1, 2.2, 4.5 and 9.6% for 21°/270° (nearly proportional; durations scale with the nominal angle, so the 270° pulse lasts 270/330 of these). The advantage is local, though: every design meets the same problem where its scan-2 pulse passes 330–350° (the margin design at $B_1^+ = 1.28$, −14%; 20°/276° at $B_1^+ = 1.25$, −14%; 17°/254° at $B_1^+ = 1.3$, +10%), so over the whole range the largest bias falls only from 20% to 10–14% for the no-crossing designs (14°/309° reaches 19.5% at its 330° point; 3% for the 180° design, whose scan-2 pulse never gets there). Off resonance the misfit of the designs with $\alpha_2 \geq 254°$ becomes detectable (flagged in 88–99% of voxels in the worst case); the bias changes by at most 2.5 points in WM and GM but grows in CSF (e.g. from 0.6 to 8.4% for 14°/309° at $B_1^+ = 1$). Finite-pulse effects are deterministic and can be built into the forward model, as is done for balanced SSFP [5]; any design whose scan-2 pulse reaches 330–360° over the B_1^+ range needs it.
- *Slab-edge spread.* All designs estimate B_1^+ as the voxel average ($\approx -5\%$) and T_1 within 2% as long as the scan-2 angle stays below about 330°. Near 360° the bias grows: −13% and −16% for 21°/270° and 20°/276° at $B_1^+ = 1.3$ (scan-2 angle 351° and 359°), and +32 to +74% where the spread straddles 360° (the published design at $B_1^+ = 1.1$–1.2, 14°/309° at 1.2), because the two sides of the null contribute signals of opposite sign. These misfits are hard to detect: the 40% CSF bias of the published design at $B_1^+ = 1.2$ is flagged in 58% of voxels, the 16% bias of 20°/276° in 10%, and the margin design's 13% in 2%. The two-stage design 17°/254°, whose scan-2 angle never exceeds 330°, stays below 1%.

2D slice profiles. An unmodelled 2D slice profile is a much larger error than the slab-edge spread and is not tabulated: the scaled small-tip profile used for the slab edge is not valid for a 250–330° sinc pulse, whose rotation axis and tip angle vary across the slice in a way only a Bloch simulation of the pulse captures. MPME relaxometry needs a 3D acquisition, as in [1], or the slice profile in the forward model.

7 Recommendations and limitations

Recommendations.

1. *Do not expect precision gains from the flip angles alone.* For the joint fit the published 15°/330° is within 2% (mean) and 3.3% (worst case) of the best T_1 bound, inside a broad band of near-optimal designs (Section 3).
2. *Keep the scan-2 pulse away from* 360° *over the whole* B_1^+ *range.* Near 360° the T_1 bound peaks (CSF), magnitude aliases come close to the truth, and unmodelled finite pulses (10–20%) and flip-angle spreads (13–75%) bias T_1, often without a detectable misfit on resonance. Crossings

of 180° are harmless. For $B_1^+ \in [0.7, 1.3]$ this means α_2 below $360°/1.3 = 277°$, and preferably with headroom:

- $\alpha_2 \approx 270°$ (e.g. 21°/270° or 18°/270°): worst-case T_1 bound within 1–2.5% of the optimum, worst-case B_1^+ bound 10–11 against 19, 33% less energy at equal pulse duration than 330°, and no aliases of any kind when fitted with $B_1^+ \in [0.667, 1.333]$ (Proposition 5), so magnitude data suffice. It pays 3% (18°/270°, all in CSF; it matches the published design in every parenchymal tissue) to 7% (21°/270°, about 60% in CSF, with 4–7% in GM and deep GM) in the mean T_1 bound;
- $\alpha_2 \approx 255°$ (e.g. 17°/254°): the scan-2 angle stays below 330° over the range, so the slab-edge spread costs $< 1\%$; it crosses 180° near $B_1^+ = 0.71$ (harmless for T_1, magnitude aliases below $B_1^+ = 0.8$) and pays about 7% in both T_1 criteria relative to the published design.

3. *Model the finite pulse duration* whatever the design: unmodelled, pulses at the peak B_1^+ of a 1 ms 330° pulse bias T_1 by 2–20%, most where the scan-2 angle passes 330–350°, which every design with $\alpha_2 \geq 254°$ does somewhere in $[0.7, 1.3]$.

4. *Two-stage pipeline*: $\alpha_1 = 17$–19° (5–7% better than 15° for the T_1 step), with $\alpha_2 \approx 255$–270° for the B_1^+ map.

5. *Under a SAR limit*, $\alpha_{\max} = 240°$ costs 6–14% and 180° costs 25–52% in the T_1 bound; below about 150° the precision falls off quickly (Fig. 5a), and a second pulse below 90° costs a factor 4.5–6.8 in the worst case (Table 3). Lower the energy by moving α_2 to 255–270° rather than below 180°.

6. *Calibrate the pulse ratio.* Every design transfers a ratio error to T_1 with a factor of at least 1.64 (1.66–3.26 over the B_1^+ range in the shortlist), undetectably.

7. *Fitting range and data type.* With a wide fitting range such as $[0.6, 1.4]$ every good design has magnitude aliases, and complex data or the FID-sign test are needed; a no-crossing design fitted within its own crossing-free range has none.

8. *Use a 3D acquisition* or model the slice profile.

Limitations. The criteria are local Cramér–Rao bounds under complex Gaussian noise; they predict the precision of an efficient estimator (which joint maximum likelihood attains for this model [2]) but not its failure rate. The design space fixes T_R, the readout and the time per scan; the robustness range weights B_1^+ uniformly and the five tissues equally, the worst case is dominated by CSF, and the worst-case optimum moves with the assumed upper limit of B_1^+. The model-error study is noise-free, uses one representative of each effect and fits the two-stage design with the joint model rather than as a two-stage pipeline. Steps that remain: Monte Carlo with the actual estimator for the shortlist (bias, efficiency, failure and alias rates over the B_1^+ range), phantom and BrainWeb simulations with a realistic B_1^+ field, and a design with the finite pulses in the forward model.

A Computation and reproducibility

Information. Single-scan Jacobians come from the exact implicit-differentiation Jacobian of the complex model (`mpme.fastjac`, 256 isochromats, float64), evaluated with a 1° nominal pulse and

$B_1^+ = x$ so that the $\ln B_1^+$ column is $\partial/\partial \ln x$ (Proposition 1). Designs: all pairs $\alpha_1 < \alpha_2$ on a 1° grid in $[1^\circ, 540^\circ]$ (145 530 pairs); B_1^+ on $[0.7, 1.3]$ in steps of 0.01, plus each pair's own crossing points $B_1^+ = 180^\circ k/\alpha_i$, where the bounds peak; five tissues. Bounds are $\sqrt{\operatorname{diag} \mathcal{I}^{-1}}$ of the diagonally normalised matrix, rescaled. Runtime about 15 minutes on a laptop CPU.

Aliases. For every pair and every true $B_1^+ \in [0.7, 1.3]$ (steps of 0.01), the roots of $f(b') - f(B_1^+)$ (complex) and $|f(b')| - |f(B_1^+)|$ (magnitude) on $b' \in [0.6, 1.4]$ are bracketed on each continuous branch of f (4001-point grid) and kept if admissible for at least one tissue ($\zeta \,|\tan(b'\alpha_1/2)/\tan(B_1^+\alpha_1/2)| < 1$). The vectorised map agrees with the exact root finder `mpme.design.alias_roots` on spot checks, and the root finder reproduces the known aliases of the published protocol at true $B_1^+ = 1$: the magnitude alias 1.1991 in $[0.6, 1.4]$ and the complex alias 2.0078 in $[0.3, 3]$.

Checks. Proposition 1 against direct evaluation of the two-scan information (`mpme.crlb`): agreement to 3×10^{-15} relative. Proposition 2 ($J_{T_1} = h(J_{B_1^+}/c - J_{M_0})$): 10^{-14} at angles from 15° to 330° and T_1 from 600 to 4000 ms. Propositions 3 and 4: Table 2. These are unit tests in `tests/test_design.py` (tolerances 10^{-10}, 10^{-12}, 10^{-9} and 10^{-8} for the four checks).

Model-error fits. Each fit of Section 6 starts from the truth and from nine points ($B_1^+ \in \{0.75, 1, 1.25\}$, $T_1 \in \{0.5, 1, 2\}\times$ truth); the lowest cost with B_1^+ in the fitting range is kept, and fits ending on a safeguard bound of the optimiser are flagged.

Scripts.

Result	Script
Bounds, biases and alias status of all designs	`scripts/flip_design.py`
Shortlist, Figs. 1 to 5, Tables 2 and 4	`scripts/flip_design_report.py`
Model-error biases, Table 5	`scripts/flip_design_robustness.py`

All are run by `make flipnote`. The design functions are in `src/mpme/design.py`.

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